

Policy-Dependent Minimal Incompatibility of Synchronous-to-Inverter Replacement Portfolios: Network-Closure Interactions and Transverse Stability

Jorge L. Mayorga Taborda

Abstract—Synchronous generators (SGs) are replaced by grid-following inverters unit by unit, yet small-signal stability is not additive: every subset of a replacement portfolio can be stable while the portfolio is not. We study the minimal incompatibility hypergraph $\mathcal{H}(\theta)$, the inclusion-minimal unstable replacement coalitions as a function of the converter control policy θ , defined exactly in transverse coordinates. The quotient removes the rotation and common-frequency drift of benchmark models without primary frequency control, and no eigenvalue is discarded by magnitude. $\mathcal{H}(\theta)$ characterizes the portfolios that are safe in every implementation order. At the descriptor level each replacement acts locally and affinely. Exact network closure between the replaced buses turns these local actions into a principal-minor interaction hierarchy of all orders, and every boundary of $\mathcal{H}$ is a zero of that hierarchy. On the IEEE 39-bus system, the reactive policy reorganizes which coalitions fail. At each of ten frozen boundaries, no interaction truncation below the cardinality of the changing minimal coalition reaches the characteristic zero. The port derivative gives the exact boundary normal and supports iterative Q/V retuning. A symmetry-deflated port extends the test to zero-frequency crossings. With documented turbine governors, the four-unit witness disappears while policy-dependent incompatibility persists in a smaller region, so $\mathcal{H}$ is a property of system and policy, not of buses. Within model-scope guards, finite-disturbance simulation reproduces $\mathcal{H}$ exactly. Preregistered Monte Carlo suites support every statement, with one disclosed rule failure.

Index Terms—Differential–algebraic equations, eigenvalue analysis, grid-following inverters, hypergraphs, low-inertia power systems, network interaction, small-signal stability, synchronous generator retirement.

I. INTRODUCTION

GENERATION portfolios are decarbonized by replacing synchronous generators (SGs) with inverter-based resources, typically plant by plant. Small-signal stability of the resulting system is a property of the *portfolio*. Low-inertia and converter-driven instabilities are part of the stability taxonomy [1]–[3]. Short-circuit ratios, single-port impedance criteria, and generalized short-circuit ratios [4]–[8] evaluate a configuration, not the combinatorial structure of a replacement plan. A planner needs a different answer: *which minimal groups of replacements are mutually incompatible, why does the network make them so, and how does the converter policy move that structure?*

J. L. Mayorga Taborda is with the Department of Electrical and Electronic Engineering, Universidad de los Andes, Bogotá, Colombia (e-mail: jl.mayorga@uniandes.edu.co).

For m candidate SGs and a converter policy θ , the minimal incompatibility hypergraph $\mathcal{H}(\theta)$ collects the inclusion-minimal unstable replacement portfolios. Its minimum edge size $\kappa(\theta)$ is the smallest incompatible coalition. Frozen benchmark models (IEEE-39, Kundur, IEEE-68) have no rotor damping and no governors, so their linearizations carry an exact double zero eigenvalue. “Stable” is therefore undefined unless the neutral directions are handled exactly.

Contributions. The paper makes three contributions; everything else is supporting evidence, robustness analysis, or a declared limitation.

- C1 Policy-dependent minimal incompatibility.** The object $\mathcal{H}(\theta)$ and $\kappa(\theta)$ are defined exactly in transverse coordinates (Theorem 1). $\mathcal{H}$ changes only at imaginary-axis crossings (Theorem 2), and a target is safe in every implementation order iff it contains no hyperedge (Proposition 2). On IEEE-39, the reactive policy reorganizes which coalitions fail; with documented governors the specific four-unit witness disappears while policy dependence persists.
- C2 Network-closure anatomy.** Replacement actions are exactly affine and bus-local in the descriptor (Lemma 1). Exact network closure converts them into principal-minor interactions of all orders, and every boundary is a zero of the interaction factor (Theorem 3, Corollary 1). At all ten frozen IEEE-39 boundaries, only the full-order interaction term of the changing coalition closes the determinant at the boundary zero; this full-order term is a closure order, not an irreducible $|S|$ -device interaction.
- C3 Actionable boundary motion.** The port derivative gives the exact local boundary normal and supports iterative Q/V retuning (Proposition 3). Symmetry deflation extends the port test to genuine zero-frequency crossings (Theorem 4).

Fig. 1 summarizes the logical chain from C2 to C1.

What is not claimed. Several ingredients are classical: the eigenvalue test (generalized Nyquist [9], [10]), Schur complements and Sylvester’s determinant identity [11], Möbius inversion [12], clutters and minimal non-faces [13], minimal failure sets in $N-k$ analysis [14], positive-systems monotonicity [15], and the NP-hardness of cardinality-constrained destabilization [16].

Three mechanistic hypotheses were tested and rejected, and none is claimed:

- the effective rank of the interaction operator explains κ ;
- a dominant network path explains a failure;
- a simple cycle causes a failure.

II. MODEL AND PROBLEM FORMULATION

A. Phasor DAE

The system is the index-one DAE

$$\dot{x} = f(x, z; \delta, \theta), \quad 0 = g(x, z; \delta, \theta), \quad (1)$$

where $x \in \mathbb{R}^n$ collects device states and $z \in \mathbb{R}^{2N}$ the rectangular bus voltages. g is Kirchhoff's current law with the bus admittance matrix, device injections, and constant-power loads. Index one means that g_z is nonsingular. At an equilibrium the reduced Jacobian is $A = f_x - f_z g_z^{-1} g_x$.

Synchronous generators. Each SG uses a two-axis model on its own base [17]. With (v_d, v_q) the terminal voltage in the rotor frame and (i_d, i_q) from the stator equations,

$$\begin{aligned} \dot{\delta} &= \omega_B(\omega - 1), & M\dot{\omega} &= P_m - P_e - D(\omega - 1), \\ T_{d0}'\dot{e}_q' &= e_{fd} - e_q' - (x_d - x_d')i_d, \\ T_{q0}'\dot{e}_d' &= -e_d' + (x_q - x_q')i_q, \end{aligned} \quad (2)$$

with $P_e = v_d i_d + v_q i_q + r_a(i_d^2 + i_q^2)$. The AVR is first order, $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$. The stabilizer is a power-input washout and lag: $v_s = K_S x_6$, $T_4 \dot{x}_6 = T_5(P_e - x_5)/T_6 - x_6$, $T_6 \dot{x}_5 = P_e - x_5$. As in the source data, $D = 0$ and P_m is constant (no governor).

Grid-following inverters. Each GFL has a synchronous-reference-frame PLL, filtered power measurements, PI outer loops, and PI inner current loops behind a filter reactance x_f :

$$\begin{aligned} \dot{v} &= k_{p,\text{pll}} v_q + x_{\text{pll}}, & \dot{x}_{\text{pll}} &= k_{i,\text{pll}} v_q, \\ \tau \dot{p}_f &= P - p_f, & \tau \dot{q}_f &= Q - q_f, \\ i_d^{\text{ref}} &= k_p(p_{\text{ref}} - p_f) + x_p, & i_q^{\text{ref}} &= -k_q(q_{\text{cmd}} - q_f) - x_q, \\ \dot{i}_{dq} &= \frac{\omega_B}{x_f} (k_i(i_{dq}^{\text{ref}} - i_{dq}) + x_{i,dq} - r_f i_{dq}), \end{aligned} \quad (3)$$

with PI states $\dot{x}_p = k_{ip}(p_{\text{ref}} - p_f)$, $\dot{x}_q = k_{iq}(q_{\text{cmd}} - q_f)$, and $\dot{x}_{i,dq} = k_{ii}(i_{dq}^{\text{ref}} - i_{dq})$. The injection is $\gamma(i_d + j i_q)e^{j\dot{\theta}}$. The reactive command comes from a leaky Q/V regulator

$$\begin{aligned} q_{\text{cmd}} &= k_{p,v} g e + x_v, & e &= v_{\text{ref}} - |V|, \\ \dot{x}_v &= k_{i,v} g e - w(x_v - q_{\text{ref}}), \end{aligned} \quad (4)$$

with leak $w = 0.05$ rad/s; $g = 0$ is exact fixed- Q operation. Parameters are listed in Table I. Converter limits, DC links, and modulation delay are not modeled.

B. Replacement portfolios and policy

Let $V = \{1, \dots, m\}$ be the candidate SGs. A portfolio $S \subseteq V$ has indicator $\delta_S \in \{0, 1\}^m$. Replacing unit i installs a GFL with the same active and reactive dispatch (*matched dispatch*) and rating equal to the SG rating, so the AC power flow is identical for every portfolio. The policy $\theta = (g, k, t, h)$ collects:

- the Q/V gain g ;
- a common scale k of every surviving SG's AVR gain;
- a scale t of every AVR time constant;

TABLE I
DEVICE AND BENCHMARK DATA

Item	Value (per unit on the device rating)
SG model	two-axis, power-form swing, $D = 0$, constant P_m
AVR	$K_A = 10.1$, $T_E = 0.25$ s (harmonized IEEEEX1)
PSS	power input: $K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,\text{pll}} = 53$, $k_{i,\text{pll}} = 1400$
Outer loops	$\tau = 0.03$ s; $k_p/k_{ip} = k_q/k_{iq} = 0.2/8$
Inner loops	$k_i/k_{ii} = 0.25/6$; $x_f = 0.15$, $r_f = 0.01$
Q/V regulator	$k_{p,v}/k_{i,v} = 2/20$; leak $w = 0.05$ rad/s; $g \in [0, 1]$
Loads	constant power
Network	IEEE 39-bus, 46 branches, 10 SGs (bus 39: interconnection)
Core	SGs 30, 33, 35, 37 (2096.6 MW; 4270.7 MVA)
Census	SGs 30–38 ($2^9 = 512$ portfolios)

- a heterogeneity exponent h , with $T_{E,i} \mapsto G(T_{E,i}/G)^h$ and G the geometric mean.

Active dispatch (MW) and converter rating (MVA) are reported separately.

III. C1: TRANSVERSE STABILITY AND THE MINIMAL INCOMPATIBILITY HYPERGRAPH

Assumption 1. (i) Every device angle state (rotor angle, PLL angle) enters (1) only through $V e^{-j \cdot}$, the other device states are written in their own rotating frames, and loads and network are rotation-covariant. (ii) Every SG has $D = 0$ and constant P_m , and the network is frequency-independent.

Let R_x equal 1 on every rotor and PLL angle and 0 elsewhere, and let w be the uniform frequency shift (speed +1, PLL integrator $+\omega_B$).

Theorem 1 (Exact transverse quotient). *Under Assumption 1(i), $AR_x = 0$. Under Assumption 1(i)–(ii), $Aw = \omega_B R_x$, so $C = \text{span}\{R_x, w\}$ is A -invariant with $A[R_x \ w] = [R_x \ w]J$, $J = \begin{bmatrix} 0 & \omega_B \\ 0 & 0 \end{bmatrix}$ (a Jordan chain). For any orthonormal Z with range $\bar{Z} = C^\perp$ and $A_\perp = Z^\top A Z$,*

$$\det(sI - A) = s^2 \det(sI - A_\perp), \quad (5)$$

and $\sigma(A_\perp)$ does not depend on Z . The reduced field $r(x) = f(x, \psi(x))$ satisfies $r(x + \phi R_x) = r(x)$ and $r(x + cw) = r(x) + c\omega_B R_x$ for all ϕ, c , so $\dot{y} = Z^\top r(x^* + Zy)$ is an exact reduced system with linearization $A_\perp$. With $D \neq 0$ or a governor, only $C = \text{span}\{R_x\}$ remains.

Proof. Rotation covariance gives $f(x + \phi R_x, e^{j\phi} z) = f(x, z)$ and a covariant g . Differentiating at $\phi = 0$ gives $f_x R_x + f_z R_z = 0$ and $g_x R_x + g_z R_z = 0$ with $R_z = jV$, so $AR_x = 0$. Under (ii), $x_0 + cw + c\omega_B t R_x$ with $z(t) = e^{j\omega_B t} z_0$ solves (1) exactly, and differentiating in c gives $Aw = \omega_B R_x$. In the basis $[U \ Z]$, with U an orthonormal basis of C , invariance gives $Z^\top A U = 0$. A is then block triangular with diagonal blocks $U^\top A U$ (spectrum $\{0, 0\}$) and $A_\perp$, which yields (5). A different complement ZO gives $O^\top A_\perp O$. The nonlinear identities are proved in Appendix A. $\square$

Definition 1. S is *transversely stable* at θ if $\alpha_\perp(S, \theta) = \max \text{Re } \sigma(A_\perp(S, \theta)) < 0$.

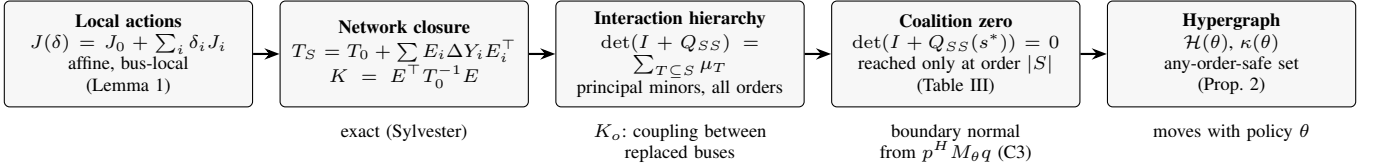


Fig. 1. The chain from local replacement actions to the minimal incompatibility hypergraph. Each arrow is an exact statement of Section IV; the IEEE-39 evidence for each box is in Section V.

The rotation is a gauge. The drift is a physical consequence of absent primary frequency control, so transverse stability is stability modulo absolute frequency restoration. The magnitude cutoff “ignore $|\lambda| \leq \varepsilon$ ” agrees with Definition 1 only when $A_\perp$ has no physical eigenvalue in that disc. In suite S1 (Section VI) it miscounts all 235 draws with a planted positive physical mode of size 10^{-5} – 5×10^{-4} , and none of the others.

Numerical labels. Every label is produced without a magnitude cutoff.

- 1) The AC equilibrium is solved to a residual below 10^{-12} , and the Jacobians are formed by central differences at steps h and $2h$, whose difference gives the error E .
- 2) R_x and w are verified ($\|AR_x\|$, $\|Aw - \omega_B R_x\|$), and $A_\perp$ and $E_\perp = Z^\top E Z$ are formed.
- 3) If $\min |\operatorname{Re} \sigma(A_\perp)| > 2 \times 10^{-3} \text{ s}^{-1}$, the label is the sign of $\alpha_\perp$.
- 4) Otherwise the label is exact if $d = \min_\omega \sigma_{\min}(A_\perp - j\omega I) > 10\varepsilon$, with $\varepsilon = \|E_\perp\| + \text{backward error}$. If not, the point is UNRESOLVED: reported, never assigned.

This is a numerical criterion, not an interval certificate.

Definition 2. For fixed θ with a transversely stable base, let $\mathcal{U}(\theta) = \{S : \alpha_\perp(S, \theta) \geq 0\}$. The *minimal incompatibility hypergraph* $\mathcal{H}(\theta)$ is the family of inclusion-minimal elements of $\mathcal{U}(\theta)$, and $\kappa(\theta) = \min_{e \in \mathcal{H}} |e|$, with $\kappa = \infty$ if $\mathcal{H} = \emptyset$.

Proposition 1 (Structure). $\mathcal{H}(\theta)$ is an antichain, and a portfolio containing no hyperedge is transversely stable. $\mathcal{U}$ need not be upward closed.

Proof. Minimal elements form an antichain. If S is unstable, a minimal unstable subset of S is a hyperedge contained in S . $\square$

For a target T , define three planning predicates:

- **A:** T is stable.
- **B:** some one-at-a-time chain $\emptyset = S_0 \subset \dots \subset S_{|T|} = T$ has every S_j stable.
- **C:** every $R \subseteq T$ is stable, so every implementation order passes only through stable stationary states.

Proposition 2 (Any-order safety). $C \Rightarrow B \Rightarrow A$, and neither converse holds in general. With a stable base, T satisfies C iff T contains no hyperedge of $\mathcal{H}(\theta)$.

Proof. C gives any chain, and B ends at T . T violates C iff some $R \subseteq T$ is unstable, iff (Proposition 1) a hyperedge lies in T . $\square$

For A without B , take diagonal vertices $A_\emptyset = -I$, $A_{\{1\}} = \operatorname{diag}(2, -3)$, $A_{\{2\}} = \operatorname{diag}(-3, 2)$, and $A_{\{1,2\}} = -2I$. For B

without C , one proper subset is unstable while another chain avoids it.

Theorem 2 (Boundary localization). Let $\theta(\tau)$ be continuous with g_z nonsingular along the path for every portfolio, and let Assumption 1 hold. Then $\mathcal{H}(\theta(\tau))$ is constant on every interval on which no $A_\perp(S, \theta(\tau))$ has an eigenvalue on the imaginary axis. Every change of $\mathcal{H}$ is bracketed by such a crossing, at finite frequency or at $s = 0$.

Proof. $A_\perp(S, \theta(\tau))$ is continuous because g_z^{-1} is, so its eigenvalues are continuous. The sign of $\alpha_\perp$ can change only through $\operatorname{Re} \lambda = 0$, and $\mathcal{H}$ is a function of the 2^m labels. $\square$

Supporting results. Two further statements are credited to known theory.

- **Monotone class.** If a common similarity makes every $T^{-1}A(S)T$ Metzler and order-preserving in S , then $\alpha_\perp$ is monotone on the lattice and $A = B = C$ [15]. One robust pair $S \subset S'$ with $\alpha_\perp(S) > 0 > \alpha_\perp(S')$ excludes every such realization.
- **Complexity.** Deciding whether some $|S| \leq k$ destabilizes is NP-complete, already for $D_\delta W D_\delta - (k-1)I$ with W a graph adjacency matrix. This is the clique reduction behind sparse PCA [16], [18]; see also [19]–[21].

IV. C2 AND C3: NETWORK CLOSURE AND BOUNDARY MOTION

A. Local actions: a descriptor realization affine in the indicators

Under matched dispatch, every candidate bus can carry *both* devices: the SG with weight $(1 - \delta_i)$ and the GFL with weight δ_i . An absent device is replaced by a stable filler $\dot{x} = -(x - x_0)$ at its intensive equilibrium, which is common to all vertices.

Lemma 1. On this realization the descriptor Jacobian is exactly affine, $J(\delta) = J_0 + \sum_i \delta_i J_i$. Each J_i is supported on the states of the devices at bus i and on the two voltage channels of bus i .

The weights multiply injections only. Each vertex has the spectrum of the actual portfolio plus the fillers’ -1 eigenvalues. The reduced $A(\delta)$ is *not* affine, because g_z depends on δ .

B. Network closure and the interaction hierarchy

Keep the network variables and eliminate only the device states: $T_S(s) = g_z + g_x(sI - f_x)^{-1}f_z$ at δ_S . Let E select

the $2m$ voltage channels of the candidate buses, and define $\Delta Y_i(s) = E_i^\top (T_{\{i\}} - T_0) E_i$ (2×2), $D = \text{blkdiag}(\Delta Y_i)$, and $K = E^\top T_0^{-1} E$. Split $K = K_d + K_o$ into its 2×2 diagonal blocks and the off-diagonal blocks, which carry the network coupling between the replaced buses. Set

$$M = DK, \quad Q = (I + DK_d)^{-1} DK_o. \quad (6)$$

Theorem 3 (Network-closure factorization). *Let s be neither a device pole nor a zero of $\det T_0$. For every S ,*

$$\det P_S(s) = h_S(s) \det T_0(s) \det(I + M_{SS}(s)), \quad (7)$$

$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \det(I + Q_{SS}), \quad (8)$$

where P_S is the descriptor pencil and $h_S = \det(sI - f_x)$ is a product of per-device factors. Moreover $\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$ with $\mu_\emptyset = 1$, $\mu_{\{i\}} = 0$, and, for $|T| \geq 2$,

$$\mu_T = \sum_{U: \text{blocks}(U)=T} \det Q_{UU}, \quad (9)$$

the sum of the principal minors of Q that touch exactly the blocks of T . μ_T is the Möbius coefficient of $S \mapsto \det(I + Q_{SS})$ and is invariant under block-diagonal changes of port basis.

Proof. The Schur complement gives $\det P_S = h_S \det T_S$. By Lemma 1, $T_S = T_0 + \sum_{i \in S} E_i \Delta Y_i E_i^\top$ exactly (Appendix A), and Sylvester's identity gives (7). Since $I + DK_d$ is block diagonal, $I + M_{SS} = (I + DK_d)_{SS} (I + Q_{SS})$, which gives (8). Multilinearity gives $\det(I + X) = \sum_U \det X_{UU}$, and grouping minors by the blocks they touch gives (9). Since $Q_{ii} = 0$, $\mu_{\{i\}} = 0$. Möbius inversion is unique [12], and a block-diagonal similarity leaves every $\det(I + Q_{SS})$ unchanged. $\square$

Corollary 1 (Boundaries are network-closure zeros). *If no member of S alone and no device has an eigenvalue at s^* , then $s^* \in \sigma(A_\perp(S))$ iff $-1 \in \sigma(M_{SS}(s^*))$ iff $\det(I + Q_{SS}(s^*)) = 0$. With $K_o = 0$ no such s^* exists.*

Every non-additive effect of a replacement portfolio is therefore an interaction term μ_T generated by network closure between the replaced buses. The algebraic degree of $S \mapsto \det(I + Q_{SS})$ and κ are different quantities: an affine family can have $\kappa = m$, and a full-degree family can have $\kappa = 1$. The theorem does not say which order closes a given boundary; that is measured in Section V.

Why aggregate indices fail. Aggregate displaced dispatch, and any other function of per-unit data, is blind to the off-diagonal blocks K_o . It therefore removes the dynamic interaction geometry contained in Q_{SS} and its principal minors, which by (8) carry the entire non-additive part of the characteristic function.

C. Boundary motion (C3)

Proposition 3 (Boundary sensitivity). *Let -1 be a simple eigenvalue of $M_{SS}(s^*, \theta^*)$ with right and left eigenvectors q, p . Then*

$$\frac{\partial s^*}{\partial \theta} = -\frac{p^H \partial_\theta M_{SS} q}{p^H \partial_s M_{SS} q}, \quad (10)$$

and $\nabla_\theta \text{Re } s^*$ is the normal of the boundary.

Proof. Simple-eigenvalue perturbation gives $d\lambda = p^H dM q / p^H q$. The implicit function theorem applied to $\lambda(s, \theta) = -1$ yields (10). $\square$

Zero-frequency crossings. By (5), $\det T_S(0)$ vanishes structurally for every S , so the port test is blind to aperiodic crossings. Let $U = R_x / \|R_x\|$ and $\beta, \beta_2 > 0$, and relocate $f_x^\# = f_x - \beta U U^\top$. The vector $x_0 = w + a R_x$ with $a = \omega_B / \beta - U^\top w / \|R_x\|$ is then a null vector of the relocated reduced matrix. Set $f_x^{\#\#} = f_x^\# - \beta_2 x_0 x_0^\top / \|x_0\|^2$ and $T^{\#\#}(0) = g_z - g_x (f_x^{\#\#})^{-1} f_z$.

Theorem 4 (Symmetry-deflated zero-frequency port). $\sigma(A^{\#\#}) = \sigma(A_\perp) \cup \{-\beta, -\beta_2\}$ and $\det T^{\#\#}(0) = \det g_z \det A^{\#\#} / \det f_x^{\#\#}$. Along a path on which $\text{sign } \det g_z$ and $\text{sign } \det f_x^{\#\#}$ are constant, $\text{sign } \det T^{\#\#}(0)$ changes iff the number of positive real eigenvalues of $A_\perp$ changes parity.

Proof. Brauer's rank-one theorem [22] moves one zero to $-\beta$. The Jordan partner becomes the exact null vector x_0 , which a second rank-one update moves to $-\beta_2$. The identity is the Schur formula. Complex pairs contribute $|\lambda|^2 > 0$, so $\text{sign } \det A_\perp = (-1)^{\dim A_\perp} (-1)^{\#\{\lambda > 0\}}$. $\square$

The test does not see complex crossings, which are covered by (7). It also does not see coalescences of two positive real eigenvalues, in which nothing crosses $s = 0$.

V. IEEE 39-BUS EVIDENCE

A. Benchmark

The IEEE 39-bus system [23] is taken from the ANDES dynamic case [24], [25], with ten two-axis SGs; bus 39 represents the interconnection and is never replaced. The documented IEEE X1 exciter is harmonized into a first-order AVR (regulator gain, exciter time constant), which is a declared model change. The core candidates are SGs 30, 33, 35, and 37; replacing all four displaces 2096.6 MW of active dispatch on 4270.7 MVA of converter rating. Every result is conditional on this model.

B. C1: policy-dependent minimal incompatibility

Three policy planes were mapped: F7A ($g \times k$), F7B ($g \times t$), and F7C ($g \times h$). All 345 229 points were re-audited in transverse coordinates on the direct path. No resolved label changed, and 1.5% of the points remain within the classifier's numerical resolution. The planes carry 30, 36, and 16 distinct hypergraphs, with up to 15 hypergraphs behind a single value of κ (Fig. 2).

- At the reference point P4 ($g = 0.03625$, $k = 1.425$, $t = 1.5$, $h = 1$), $\mathcal{H} = \{\{30, 33, 35, 37\}\}$ and $\alpha_\perp = +0.127 \text{ s}^{-1}$: every proper subset is stable and the four-unit portfolio is not.
- At full voltage-support gain (P_∞ : $g = 1$, $k = 0.5$), $\mathcal{H} = \emptyset$.
- On the line $t = 0.852$, the minimal witness contracts $4 \rightarrow 3 \rightarrow 2$ and re-expands $3 \rightarrow 4 \rightarrow \emptyset$ as g increases (Fig. 6).

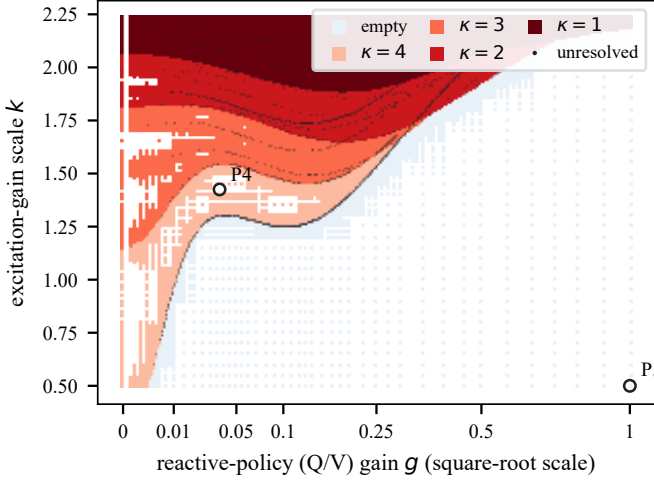


Fig. 2. Transverse minimum incompatible order κ over the F7A policy plane of IEEE-39 ($t = 1.5$, $h = 1$, core 30/33/35/37, matched dispatch, no governor). Black dots: points within classifier resolution.

TABLE II
FROZEN VERSUS GOVERNED IEEE-39 (DOCUMENTED TGOV1N)

Quantity	Frozen	Governed
P4 four-unit $\alpha_{\perp}$	+0.127	-0.0745
P4 line, $\kappa = 4$ window	$g \in [0.016, 0.20]$	$g \in [0.016, 0.023]$
tongue line $k = 1.30$	U-S-U-S (re-entry)	no re-entry
plane: distinct $\mathcal{H}$	26	10
plane: non-empty points	148 (+21 base unstable)	74 (base always stable)
plane: $\kappa = 1$ points	48	0
reference points	5 of 6	1 of 6 (P3)
non-empty		
damped condenser, P4	threshold 2.48–3%	not needed

Primary frequency control. The same source workbook documents TGOV1N turbine governors ($R = 0.05$, $T_1 = 0.05$, $T_2 = 1$, $T_3 = 2.1$ s). They were added untuned as a separately versioned model, for which $C = \text{span}\{R_x\}$ (Theorem 1). Fig. 3 and Table II show the result:

- The P4 four-unit portfolio becomes stable ($\alpha_{\perp} = -0.0745$), although it remains the least-damped portfolio of the lattice.
- On a 399-point plane the number of distinct hypergraphs falls from 26 to 10. Non-empty structure remains at 74 points ($\kappa = 2, 3, 4$), no singleton fails, and the base is always stable.
- The re-entrant tongue on $k = 1.30$ disappears.
- In ANDES the governors move the base inter-area pair from -0.222 to -0.770 s^{-1} , which is consistent in direction.

The specific four-unit witness is conditional on the absence of primary frequency control, while policy-dependent incompatibility persists in the governed model in a smaller region. $\mathcal{H}$ is a property of the system and policy, not a permanent label on buses.

Displaced MW is not the variable. The failing mode is the 0.64–0.71 Hz inter-area family (Fig. 4). The 25 stable four-unit census portfolios whose displaced dispatch matches that

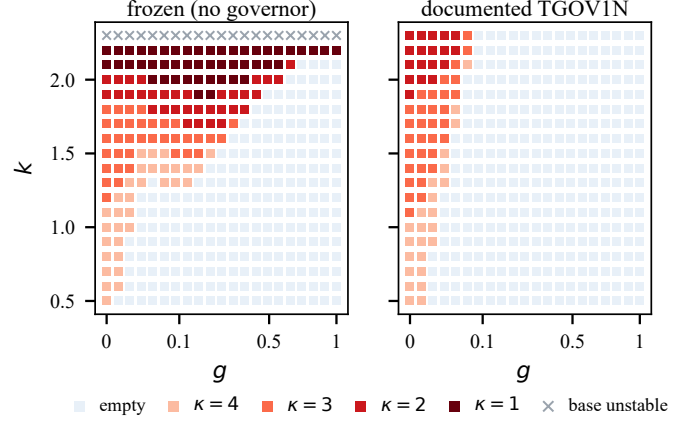


Fig. 3. Minimum incompatible order on a coarse IEEE-39 policy plane ($t = 1.5$, $h = 1$): frozen model versus the model with the source's documented TGOV1N governors.

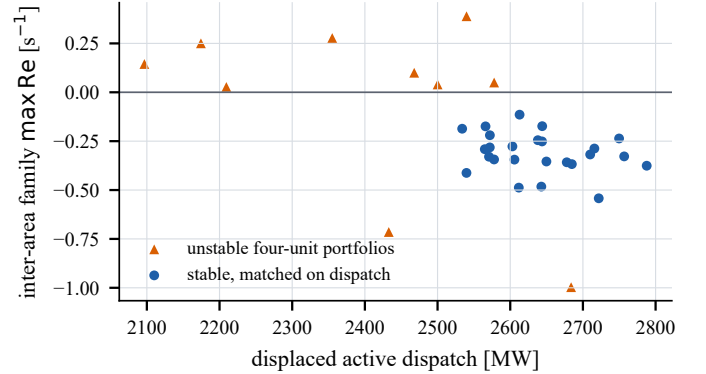


Fig. 4. Inter-area family abscissa versus displaced active dispatch for the unstable four-unit census portfolios and 25 stable portfolios matched on dispatch.

of the 10 unstable ones all keep the family damped ($\max \text{Re} \leq -0.114$ s^{-1} , $\zeta \geq 3.9\%$). The family is unstable in 8 of the 10 failing portfolios, and total displaced dispatch predicts the minimal coalition with AUC 0.57. Section IV explains why: MW is blind to the interaction geometry in Q_{SS} .

C. C2: network-closure anatomy of ten frozen boundaries

Ten frozen boundaries were re-located by bisection and analyzed with Theorem 3: the nine witness changes of the F7B line $t = 0.852$, and the four-unit boundary on the P4 line ($g^* = 0.2077$, 0.706 Hz).

- *Local actions.* The descriptor blocks f_x, f_z, g_x, g_z are affine in δ to 4.5×10^{-16} (order- ≥ 2 Möbius coefficients). Each replacement touches only its own devices and bus.
- *Exact closure.* (7) holds to 1.4×10^{-13} for all 16 subsets at all ten points, and the critical eigenvalue of M_{SS} equals -1 to 8.5×10^{-14} .
- *Local factor never vanishes.* $\prod_i \det(I + M_{ii})$ stays between 0.08 and 0.31. The zero is carried entirely by $\det(I + Q_{SS})$.

The central result is the following observation (Table III, Fig. 5). *For every one of the ten frozen IEEE-39 boundaries,*

TABLE III
NETWORK-CLOSURE ANATOMY OF TEN FROZEN IEEE-39 BOUNDARIES

Ev.	S changing	g^*	Hz	local	trunc. ^a	$\sigma_1(Q)$	rank ^b	angle ^c
E1	30+33+35	0.0102	0.628	0.315	0.336	6.13	7	2.3e-4
E2	30+33+37	0.0188	0.633	0.198	0.396	6.88	7	7.1e-5
E3	33+35+37	0.0609	0.671	0.080	0.385	4.00	8	2.7e-4
E4	30+33	0.1034	0.656	0.081	1	8.93	7	1.8e-4
E5	33+35+37	0.1870	0.695	0.135	0.296	2.37	8	3.3e-4
E6	30+33	0.2491	0.672	0.104	1	6.06	7	8.4e-5
E7	30+33+35	0.2778	0.706	0.130	0.372	2.01	8	3.5e-4
E8	30+33+37	0.2931	0.685	0.132	0.293	3.46	8	2.0e-4
E9	30+33+35+37	0.2973	0.718	0.134	0.132	1.67	8	2.9e-4
F	30+33+35+37	0.2077	0.706	0.089	0.161	1.84	8	4.3e-4

E1–E9: F7B line $t = 0.852$ ($k = h = 1$); F: P4 line ($k = 1.425$, $t = 1.5$).
^a $|\det(I + Q_{SS})|$ truncated at order $|S| - 1$ (for pairs this is order 1, which is identically 1); the full order gives $\leq 4 \times 10^{-13}$. ^b Effective rank of Q at 10^{-2} . ^c Boundary-normal error of (10) against direct finite differences [deg].

no interaction truncation below the cardinality of the changing minimal coalition reaches the characteristic zero. Truncating at order $|S| - 1$ leaves 0.29–0.40 for the triples and 0.13–0.16 for the four-unit portfolio, whereas the full order reaches $\leq 4 \times 10^{-13}$. Table IV shows the build-up at the four-unit boundary. The interaction coefficients of orders 2, 3, and 4 have comparable magnitude (0.51, 0.45, and 0.16). The fourth-order term, a sum of principal minors of Q that touch all four replaced buses, is exactly what closes the determinant. Across 120 random (θ, s) draws (suite P3), the full-order coefficient never vanishes (7.8×10^{-4} to 469), so no lower-degree representation is exact on this benchmark.

Failing-set cardinality is not interaction order. At the flagship operating point, $\kappa = 4$ denotes the cardinality of the minimum failing replacement portfolio: all proper subsets remain stable while the complete four-unit portfolio is unstable. This should not be interpreted as evidence of an irreducible four-device interaction. The exact full-support Boolean closure contribution is required at the observed boundary, but the connected-cumulant decomposition shows that this contribution is composite, containing the connected four-way term together with products of lower-order connected terms. Let χ_T be the connected (partition-lattice) cumulants of $S \mapsto \det(I + Q_{SS})$ [12], [13]. Because Q has zero diagonal blocks, $\det(I + Q_{ii}) = 1$ and μ_T is the sum of $\prod_{B \in \pi} \chi_B$ over the set partitions π of T without singleton blocks; for four units

$$\mu_{1234} = \chi_{1234} + \chi_{12}\chi_{34} + \chi_{13}\chi_{24} + \chi_{14}\chi_{23},$$

an identity checked numerically to about 3×10^{-14} . At the four-unit boundary on the P4 line, deleting $\chi_{\{30,33,35,37\}}$ moves the characteristic zero by only 0.024 s^{-1} , and the connected cluster with the largest effect on the zero is $\{33, 35\}$. The closure statement above therefore concerns the order of the determinant expansion at the boundary zero, not an irreducible interaction order. Nor does it mean that no lower-order computation can flag the portfolio: at P4 itself the order- ≤ 3 truncation of $\det(I + Q_{SS})$ has its zero at $+0.1267 \text{ s}^{-1}$ (exact $+0.1270 \text{ s}^{-1}$), whereas additive, pairwise, and third-order truncations of the spectral abscissa all predict stability (-0.18 to -0.21 s^{-1}). The cumulant decomposition is used only for this interpretation; it is not a contribution of the paper.

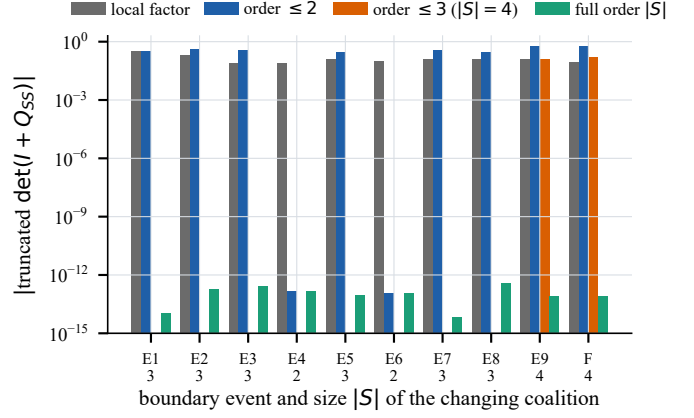


Fig. 5. Anatomy of the ten frozen IEEE-39 boundaries (E1–E9: F7B line $t = 0.852$; F: P4 line). Shown are the local factor and the truncations of the interaction expansion (9). Only the full-order term (order $|S|$) reaches the characteristic zero. At the four-unit boundary this term contains products of pair interactions besides the connected four-device term, so the full order is a closure order, not an irreducible interaction order. For pairs, order ≤ 2 is the full order.

TABLE IV
ORDER-BY-ORDER EXPANSION AT THE FOUR-UNIT BOUNDARY (P4 LINE, 0.706 Hz)

Order k	$\det(I + M_{SS})$		$\det(I + Q_{SS})$	
	$ \sum_{ T =k} \mu_T $	$ \text{trunc.} \leq k $	$ \sum_{ T =k} \tilde{\mu}_T $	$ \text{trunc.} \leq k $
0	1	1	1	1
1	2.509	1.712	0	1
2	2.393	0.889	0.509	0.605
3	0.989	0.146	0.445	0.161
4	0.146	7×10^{-15}	0.161	8×10^{-14}

The left pair includes the local terms; in the right pair they are factored out ($\tilde{\mu}_{\{i\}} = 0$, Theorem 3).

Negative control: interaction dimension. The effective rank of Q (threshold 10^{-2}) stays at 7–8 of 8 while κ changes $4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4$ (Fig. 6). Interaction dimension is therefore not the mechanism of witness contraction, and we do not claim that rank explains κ . The magnitude $\sigma_1(Q)$ varies (6.1–8.9 at the contracting events, 1.7–1.8 at the four-unit boundaries), but this is only descriptive.

Limitation: no path attribution. A bus-diagonal Neumann walk expansion of T_0^{-1} ($T_0 = D_g - N$) does not converge in the inter-area band ($\rho(D_g^{-1}N) \geq 1.003$ over 0.3–1.5 Hz), both at the four-unit boundary and for a Pg-matched stable control. No n-hop attribution is claimed. The closure result concerns the closure order of the interaction expansion, not a dominant path or cycle; the simple-cycle explanation was tested separately and falsified.

D. C3: boundary motion, retuning, and zero-frequency crossings

Formula (10) was compared with direct finite differences of the tracked transverse eigenvalue in all four policy coordinates. At all ten boundaries the boundary normal agrees to 4.3×10^{-4} degrees, its magnitude to 4.3×10^{-6} (relative), and the frequency drift to 9×10^{-6} (Table III).

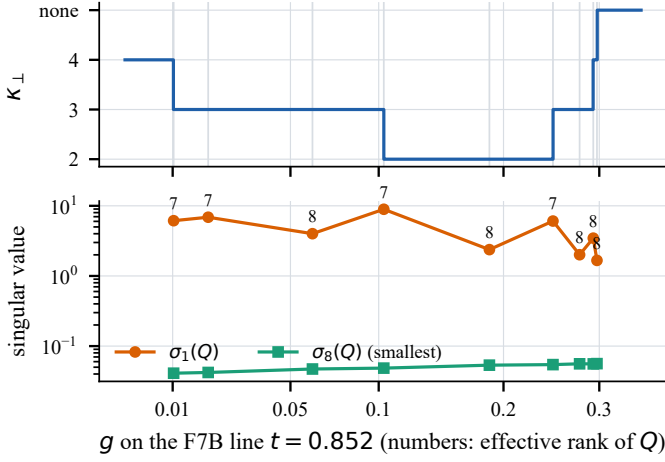


Fig. 6. F7B policy line $t = 0.852$. Top: the transverse minimum incompatible order contracts $4 \rightarrow 3 \rightarrow 2$ and re-expands. Bottom: largest and smallest singular values of Q at the nine boundary events, labelled with the effective rank. The dimension stays full while κ changes (negative control).

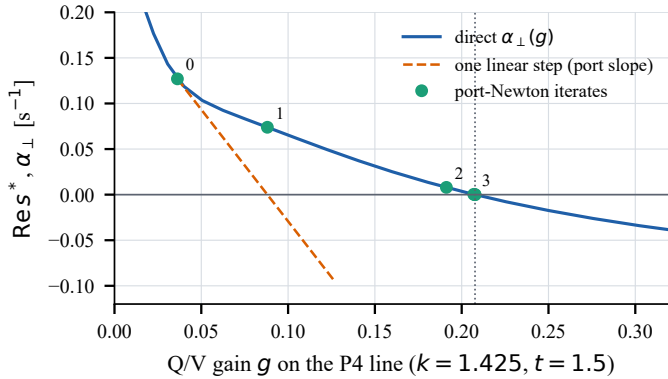


Fig. 7. Q/V retuning of the P4 four-unit portfolio. The single linear step built from the port derivative (10) undershoots because the slope changes fivefold along the path. Newton iteration with the same derivative reaches the continuation boundary (dotted) to 2×10^{-10} .

Consider making the P4 portfolio stable by a pure Q/V-gain change (Fig. 7). A single linear step predicts $\Delta g = 0.052$, whereas continuation gives 0.171 : the slope $\partial \text{Re } s / \partial g$ drifts from -2.45 at P4 to -0.46 at the boundary. Newton iteration with (10) reaches the continuation boundary $g^* = 0.20768$ to 2×10^{-10} in five steps. The port derivative is therefore an exact local tool and an iterative retuning tool, not a one-shot predictor.

IEEE-39 has no real transverse crossing, so Theorem 4 is exercised on the Kundur two-area system. There, all 194 aperiodic crossings of the policy maps are leaky Q/V-integrator modes passing through $s = 0$. These 194 motivated the construction and are not counted as validation. The frozen construction was then evaluated on held-out Kundur slices (Fig. 8):

- actual zero-frequency crossings: 28, detected 28/28;
- one further real-count change is a coalescence of two positive real eigenvalues into a complex pair, correctly not a zero-crossing event;
- false positives on 445 controls (109 oscillatory crossings,

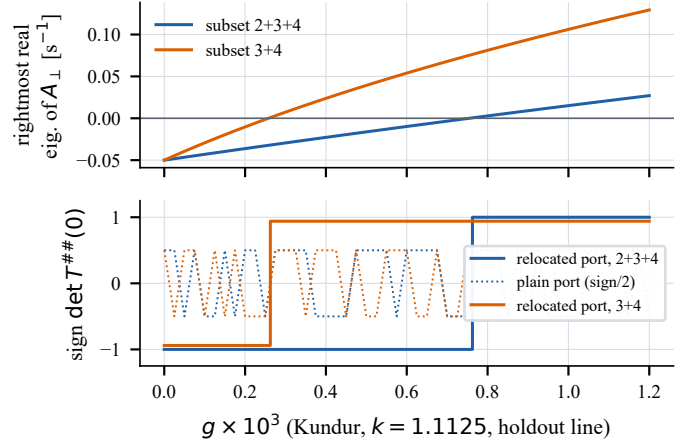


Fig. 8. Symmetry-deflated zero-frequency port on a Kundur holdout line. Top: the physical real transverse eigenvalue crosses zero. Bottom: the relocated port changes sign exactly at the crossings; the plain port (dotted) is structurally singular and its sign is numerical noise.

TABLE V
PLANNING RESULTS ON IEEE-39 (MW AND MVA ON SEPARATE AXES)

Problem	Setting	Result
Any-order safe	core, P4	30+33+35: 1775.1 MW / 3300.5 MVA
Any-order safe	core, P_∞	30+33+35+37: 2096.6 MW / 4270.7 MVA
Any-order safe	9-unit census	31+32+34+35+37+38: 3652.3 MW / 6499.9 MVA
Order dependence	9-unit census	3 of 327 stable targets
Nonlinear safe	core, P4	out of model scope
Nonlinear safe	core, P_∞ , D2	identical to spectral
Support	P4, small signal	106.0 MVA (damped condenser)
Support	P4, envelope	684–792 MVA (not a stability requirement)
Governors	P4	stable without support

336 no-crossing brackets): 0.

E. Planning consequences

Proposition 2 is exact, so $\mathcal{H}$ becomes a set-packing constraint for any-order-safe portfolios (Table V).

- Order dependence exists but is rare. On the 512-portfolio census at one reference policy, 327 portfolios are stable and 3 of these are order-dependent; each becomes unsafe exactly when unit 34 is converted last. On the P2 random paths, 131 of 30 755 stable targets are order-dependent.
- The three census pairs also exclude the monotone class.
- The MW-optimal census plan (3652.3 MW / 6499.9 MVA) is safe in any order.

Sequencing is therefore not the dominant application. The main planning use is the any-order-safe set and its movement with policy.

A damped condenser restores P4 above 2.48% of the retired rating (106.0 MVA). Keeping a 200-MW disturbance inside the declared converter voltage band requires 684–792 MVA; this is an envelope requirement of the model, not a stability requirement.

VI. MONTE CARLO VALIDATION (SUPPORTING)

All pass rules were frozen in a configuration file committed before any draw. Synthetic suites S1–S7 draw random systems that satisfy exactly one statement’s hypotheses (Appendix C). The IEEE-39 suites P1–P4 draw random policies ($g = u^2$, $u \sim U(0, 1)$; $k \in [0.5, 2.3]$; $t \in [0.5, 3]$; $h \in [0, 2]$) and random portfolios of the core:

- P1 checks Theorem 1, including the nonlinear symmetries at random off-equilibrium states;
- P2 follows random policy segments and re-brackets every change of $\mathcal{H}$;
- P3 checks (7) and (9) at random s ;
- P4 compares nonlinear TDS (5-MW pulse) with the sign of $\alpha_\perp$ on 40 stable and 40 unstable draws.

Table VI and Fig. 9 report the results. Checks that are logical identities validate the implementation; the informative outputs are the physical checks and the frequencies.

All preregistered IEEE-39 checks passed except the P2 localization tolerance rule, which failed as implemented; its post-hoc re-bisection is reported separately. The P2 bisection used the three-state label and therefore stopped at the edge of the classifier’s unresolved band. There the crossing eigenvalue still has $|\operatorname{Re} \lambda| = 4 \times 10^{-6} - 7 \times 10^{-5}$, above the frozen 10^{-6} , so the rule fails as preregistered. Separately, and post hoc, a re-bisection on the sign of $\alpha_\perp$ (same paths, brackets, and threshold) locates all 120 events at $|\operatorname{Re} \lambda| \leq 2 \times 10^{-9}$. Each is a complex pair at 0.41–0.72 Hz with finite $|\lambda|$ and g_z well conditioned along every path, as Theorem 2 requires. The corrected result is not counted as a preregistered pass. The TDS envelope rate tracks $\alpha_\perp$ ($r = 0.969$), which confirms that Definition 1 is the stability notion the nonlinear model obeys. The synthetic frequencies show that the converses of Proposition 2 fail often in random families (65% have a stable target that is not safe in every order, and 4% have one that is unreachable by any safe path), and that the monotone class is narrow (99.7% of sign-unconstrained families violate it).

Deviations from the preregistration.

- 1) The P1 spectral-identity rule matched the numerically split Jordan pair ($\mathcal{O}(\sqrt{\epsilon} \|A\|_{\omega_B}) \approx 10^{-5}$). It was amended before the run, with the threshold unchanged.
- 2) The P2 rule failed as implemented (above).
- 3) In S7, three coalescence-class draws contained a genuine zero crossing. In addition, 183 draws were left undecided by the device-flip rule, because synthetic device blocks are not block diagonal; no Kundur holdout bracket was undecided.
- 4) Two P1 draws needed a redrawn off-equilibrium state that stayed inside the algebraic chart.

VII. SCOPE CHECKS AND LIMITATIONS

A. Nonlinear finite disturbances: a negative result

Nonlinear composability is a scope check, not a contribution. Three frozen disturbance families were used: a bus-16 load pulse (D1), an SG36 mechanical-power dip (D2), and a GFL output dip (D3). Model-scope guards were declared beforehand (Appendix D). Inside these guards, the

finite-disturbance hypergraph equals the spectral $\mathcal{H}(\theta)$ in every tested case (six point-family combinations; Fig. 10). No transversely stable portfolio fails inside the guards: 84 of 96 thresholds are guard activations (GFL voltage band, stabilizer limit). Earlier preregistered runs confirmed all 32 declared small-disturbance verdicts, and P4 extends this to 80 random cases. The three examined oscillatory boundaries are subcritical Hopf points [26], with first Lyapunov coefficients $+0.0095$, $+0.025$, and $+0.026$. Beyond the guards the model does not decide.

B. Limitations

- *Model.* The model is a phasor DAE, not an electromagnetic transient model. It has no converter current limits, ride-through logic, or DC link, and the IEEE X1 exciter is harmonized. The frozen models have no primary frequency restoration, so all stability statements are transverse. The documented governors change the P4 conclusion (Table II).
- *Witness identity.* Under the preregistered parameter-uncertainty envelopes, portfolio non-composability persisted, but the identity of the nominal minimum failing set was parameter-sensitive. Non-composability held in 88%, 84%, and 100% of the per-unit machine, combined, and converter-stress draws (62% fleet-wide), whereas the nominal witness $\{30, 33, 35, 37\}$ was retained in 34% of the fleet-wide, 69% of the per-unit, and 62% of the combined draws (coverage fractions of deterministic envelopes, not probabilities). Accordingly, $\{30, 33, 35, 37\}$ is reported as the witness of the nominal frozen model, not as a robust weak-bus set.
- *Independent implementation.* The network and operating point were reproduced in ANDES 2.0.0 and, after a tap-convention translation of the three transformers whose off-nominal tap is on the low-voltage from-bus, in pandapower 3.4.0 [27] (bus-admittance matrices identical to 10^{-13} p.u.; bus voltages within 2×10^{-7} p.u., the residual being ANDES’s internal 10^{-8} p.u. branch-impedance regularization). On an equation-equivalent synchronous configuration ANDES reproduces the electromechanical eigenvalues within $2 \times 10^{-6} \text{ s}^{-1}$ and the branch sensitivities of the critical mode on a preregistered 12-branch holdout (12/12 signs, Spearman 1.00). With the two-axis machine and the grid-following converter of this paper transcribed into ANDES as custom models, ANDES reproduces all 32 transverse verdicts of the 16 subsets of $\{30, 33, 35, 37\}$ at P4 and at $g = 0.25$ ($|\Delta \alpha_\perp| \leq 1.3 \times 10^{-6} \text{ s}^{-1}$), including $\mathcal{H} = \{\{30, 33, 35, 37\}\}$ and $\kappa = 4$ at P4. This independently reproduces the computation, not the adequacy of the converter model: no conclusion is claimed for other converter models, and with a library converter model the branch ranking changes (7/12 signs). ANDES also confirms the governor damping direction.
- *EMT corroboration.* EMT corroboration was preregistered in ParaEMT [28]. Although the IEEE-39 network and operating point were independently reproduced, the campaigns terminated at pre-specified device-interface

TABLE VI
PREREGISTERED MONTE CARLO VALIDATION

Statement	Suite	Draws	N	Measured	Frozen rule / result
Thm 1 (linear)	S1 synthetic	1000	1000	exact RHP count 1000/1000; (5) $\leq 5.5 \times 10^{-9}$; basis 1.5×10^{-16}	$\leq 10^{-8}$: pass
control: magnitude cutoff	S1	1000	1000	wrong in 235/235 positive planted modes, 0/765 otherwise	expected failure: confirmed
Props 1,2	S2 synthetic	2000	30 881 targets	0 violations; $A \setminus B$ in 86 families, $B \setminus C$ in 1276	0 violations: pass
Thm 2	S3 synthetic	300	1044 crossings	every $\mathcal{H}$ change bracketed (698 real, 346 complex)	0 exceptions: pass
monotone class	S4 synthetic	1000	80 000 edges	0 non-monotone edges, $A=B=C$	pass
control: signs removed	S4	1000	80 000	37 166 non-monotone edges in 997/1000 families	expected failure: confirmed
complexity	S5 synthetic	500	exhaustive	0 mismatches with k -cliques	pass
Thm 3 (algebra)	S6 synthetic	500	500	minors 3.4×10^{-15} , Möbius 1.3×10^{-15} , invariance 8.2×10^{-12}	$\leq 10^{-10}$: pass
Thm 4	S7 synthetic	1000	817 decided	flip $\Leftrightarrow$ parity change in 817/817; 183 undecided by the device-flip rule	pass ^a
Thm 1 (physical)	P1 IEEE-39	300	300	$\dim C = 2$ in 300/300; spectrum 4.5×10^{-9} ; Jordan entry 3.9×10^{-12}	pass
Thm 1 (nonlinear)	P1 IEEE-39	300	300	$r(x + \phi R_x)$: 2.4×10^{-14} ; $r(x + cw)$: 9.4×10^{-15}	$\leq 10^{-8}$: pass
Thm 2, Prop 2	P2 IEEE-39	120 paths	120 events	101 changes of $\mathcal{H}$ on 28 paths; all crossings complex, 0.41–0.72 Hz; $\text{cond}(g_z) \leq 2.8 \times 10^3$	fail as implemented^c (post hoc, reported separately: 120/120)
Thm 3 (physical)	P3 IEEE-39	40	120	ratio 3.2×10^{-13} ; Möbius 8.1×10^{-13}	$\leq 10^{-8}$: pass
Def. 1 in TDS	P4 IEEE-39	80	80	TDS verdict = sign of $\alpha_\perp$ in 80/80; rate vs. $\alpha_\perp$: $r = 0.969$	80/80: pass
Thm 4 (physical)	Kundur holdout	474	28+445	zero crossings detected 28/28; coalescence not flagged; 0/445 false positives	pass ^b

^aThree draws of the “coalescence” class contained a genuine zero crossing (generator defect); the port flipped in exactly those. ^bThe preregistered tally counted 29 real-count changes; one is a coalescence of two positive real eigenvalues into a complex pair, in which no eigenvalue crosses $s = 0$, so it lies outside the scope of Theorem 4. ^cThe bisection used the three-state label and stopped at the classifier’s resolution band ($|\text{Re } \lambda| \leq 7 \times 10^{-5} > 10^{-6}$). The separate post-hoc re-bisection on the sign of $\alpha_\perp$ (threshold unchanged) gives $\leq 2 \times 10^{-9}$ and is not counted as a preregistered pass.

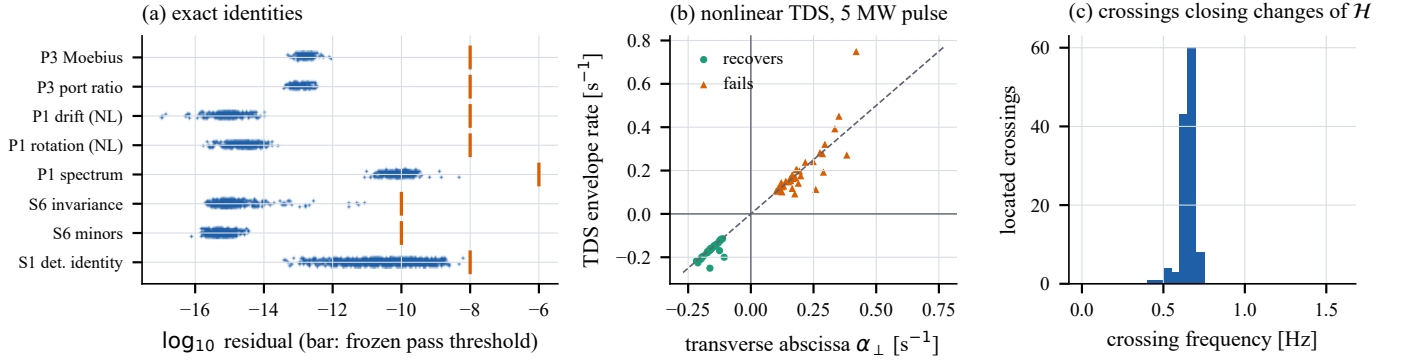


Fig. 9. Monte Carlo validation. (a) Residuals of the exact identities (S1, S6 synthetic; P1, P3 IEEE-39) against their frozen thresholds (bars). (b) Nonlinear TDS envelope growth rate versus the transverse abscissa for 80 random IEEE-39 policies and portfolios (dashed: identity). (c) Frequencies of the transverse eigenvalues that close the 120 changes of $\mathcal{H}$ on random policy paths (post-hoc re-bisection; Table VI, note c).

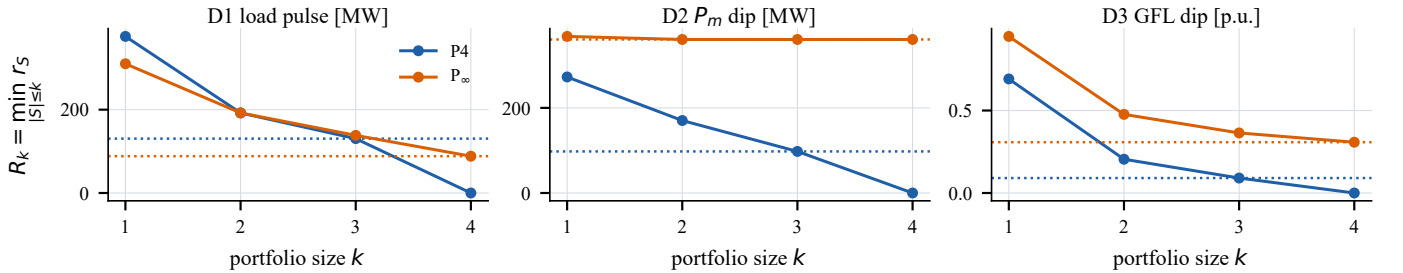


Fig. 10. Finite-disturbance resilience radii $R_k = \min_{|S| \leq k} r_S$ of the IEEE-39 core at P4 and P_∞ (TDS, phasor DAE). Dotted: model-scope radius ρ_{scope} . Below it the nonlinear hypergraph equals $\mathcal{H}(\theta)$; above it every threshold is a guard activation.

and modal-identification qualification gates before any portfolio-level experiment; therefore, no EMT validation of the portfolio claims is asserted. A methodological observation from the modal-identification qualification (linear modal residues of the phasor models; not an EMT result) is kept outside the claims: the dominant observed modal energy need not correspond to the stability-limiting eigenmode (in 10 of 26 qualification models the rightmost in-band mode carried a minority of the channel energy).

- *System dependence.* On the documented IEEE 68-bus system [29], which is reproduced to 15/15 modes within 5×10^{-4} Hz, the preregistered four-candidate policy map is empty. Policy dependence appears only at ≥ 6 of 12 replaced plants. On the Kundur system, 61/61 and 36/36 policy lines change $\mathcal{H}$.
- *Mechanism scope.* No rank, path, or cycle explanation is claimed (Section V-C). The Neumann walk expansion is non-executable in the relevant band.
- *Numerics.* Labels are numerically exact, not interval-certified, and 1.5% of policy-map points are unresolved. No stability certificate is claimed: sampled small-gain screens give $\rho = 2.9\text{--}12.3 > 1$, and a local Lyapunov bound reaches $\leq 10^{-11}$ of the measured threshold.
- *Scalability.* Enumerating 2^m labels is exponential, and minimum-cardinality destabilization is NP-hard. Equation (7) evaluates every portfolio from one base network and m local increments, but it does not remove the combinatorics.

VIII. DISCUSSION

Table VII places the work among established method families. The hypergraph does not replace any of them, and its per-portfolio labels *are* generalized Nyquist tests. What is added is the organization of those labels over binary replacement decisions, their network-closure anatomy, and their movement with the converter policy. Flagship-only Nyquist, single-port impedance, and short-circuit-ratio screens miss the coalition changes on IEEE-39. Modal-sensitivity, additive, and pairwise predictors of the four-unit instability were correct in 22, 31, and 38 of 52 cases, respectively (third order: 50).

IX. CONCLUSION

The minimal incompatibility hypergraph $\mathcal{H}(\theta)$ is defined exactly in transverse coordinates, changes only at imaginary-axis crossings, and characterizes the portfolios that are safe in every implementation order (C1). Its non-additivity is the principal-minor hierarchy produced by network closure of local, affine replacement actions. At each of ten frozen IEEE-39 boundaries, no truncation below the cardinality of the changing minimal coalition (a closure order, not an irreducible interaction order) reaches the characteristic zero (C2). The port derivative gives exact boundary normals and iterative retuning, and symmetry deflation extends the test to zero-frequency crossings (C3).

The reactive policy reorganizes the structure. The specific four-unit witness is conditional on the absence of primary

TABLE VII
RELATION TO ESTABLISHED METHOD FAMILIES

Family	Relation to this work
Generalized Nyquist [10]	per-portfolio test; our labels are GN tests
Impedance / SCR [4], [7]	configuration screens; miss coalition changes
Small gain, passivity μ / robust stability [19]	sampled small-gain screen fails ($\rho = 2.9\text{--}12.3$) NP-hard in general; no priority claimed
Sparse stability radius [21]	continuous perturbations, not binary replacement
$N-k$ minimal sets [14]	static analogue; ours is dynamic, transverse, and policy dependent, with network-closure anatomy
Selective modal analysis [30]	explains modes; does not enumerate coalitions
Continuation / Hopf [26]	boundaries are subcritical Hopf points
Energy / Lyapunov	local bound $\leq 10^{-11}$ of the threshold
Higher-order modal [31]	requires $f_z D^2 \psi$ (Appendix B)

frequency control, while policy dependence persists with documented governors. Within model scope, nonlinear simulation reproduces $\mathcal{H}$ exactly. Neither interaction rank nor network paths explain the witness order. Every statement was tested by preregistered Monte Carlo suites; one rule failed as implemented and is disclosed with its separate post-hoc correction. Limiter-equipped, governed, and mixed grid-forming benchmarks require new, separately versioned models.

APPENDIX A PROOF DETAILS

Nonlinear identities in Theorem 1. Fix x in the chart domain, let $z = \psi(x)$, and set $x_c = x + cw$. Speeds enter the reduced equations only through $\dot{\delta} = \omega_B(\omega - 1)$ and, for a GFL, through $\dot{\vartheta} = k_{p,\text{pll}}v_q + x_{\text{pll}}$. In both, the shift adds the constant $c\omega_B$ to the angle derivatives and nothing else. The power-form swing equation with $D = 0$ does not contain ω on its right-hand side, the stabilizer uses electrical power, and the network is frequency independent. Hence $\psi(x_c) = \psi(x)$ and $r(x_c) = r(x) + c\omega_B R_x$. The rotation identity follows from $\psi(x + \phi R_x) = e^{j\phi} \psi(x)$ and the invariance of f under a common rotation of angles and voltages.

Locality in Theorem 3. On the common realization f_x is block diagonal by device. f_z and g_x couple a device only to the two channels of its own bus, and the device contribution to g_z is a 2×2 block on that bus. Hence $g_x(sI - f_x)^{-1}f_z = \sum_{\text{devices}} E_b G_b(s) E_b^\top$, where G_b is the device's 2×2 admittance. With network and load terms common to all vertices, $T_S - T_0 = \sum_{i \in S} E_i \Delta Y_i E_i^\top$. ΔY_i does not depend on S because every device stays at its common intensive equilibrium. On the direct (non-common) path the locality residual is bounded by the power-flow tolerance and was measured below 8×10^{-8} .

Corollary 1. If member i alone has no eigenvalue at s^* , then $\det(I + M_{ii}(s^*)) = \det T_{\{i\}}(s^*) / \det T_0(s^*) \neq 0$, so the zero of (8) is a zero of $\det(I + Q_{SS})$. With $K_o = 0$, $\det(I + Q_{SS}) = 1$.

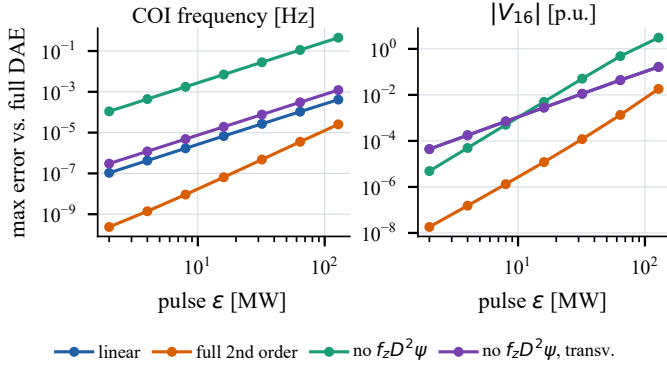


Fig. 11. Second-order reduced response of the triple 30+33+35 at P4 against the full DAE. The complete second-order model requires $f_z D^2 \psi$.

APPENDIX B

SECOND-ORDER CURVATURE AUDIT (SUPPORTING METHODOLOGY)

On the same DAE, three reduced models of the response to a bus-16 load pulse were compared with the full nonlinear solution (Fig. 11):

- linear;
- complete second order;
- second order without the algebraic curvature $f_z D^2 \psi$.

At 2 MW the complete second-order model is 170–450 times more accurate than the linear one in frequency and 750–3400 times in bus voltage (slope 3.0–3.2). Dropping $f_z D^2 \psi$ breaks the rotation/drift symmetry at second order and makes the frequency prediction 570–1700 times worse than linear. With the center-subspace component removed, the device curvature alone is still no better than linear. This is a consistency requirement of second-order reduction [31], not a novelty.

APPENDIX C

MONTÉ CARLO CONSTRUCTIONS

S1. $n \sim U\{8, \dots, 40\}$. The transverse spectrum has real parts in $[-2, 1]$ (with $|\text{Re}| \geq 0.01$) and imaginary parts in $[0.5, 10]$. Odd draws plant one real eigenvalue at $\pm u$, with u log-uniform in $[10^{-5}, 5 \times 10^{-4}]$. $P = Q(I + 0.3G)$ with Q Haar orthogonal. *S2.* $A(\delta) = A_0 + \sum \delta_i A_i$ with $m \in \{4, 5, 6\}$, $n = 10$, $\alpha(A_0) \in [-1, -0.2]$, and rank-2 actions; lattice search is exhaustive. *S3.* Actions $A_i + pB_i$ on a 401-point grid; 50-step bisection. *S4.* $B(S) = B_0 + \sum_{i \in S} N_i$, B_0 Metzler, $N_i \geq 0$, under random similarity ($\text{cond} \leq 10^3$); the control uses random signs. *S5.* $G(n, p)$ with $n \in [7, 10]$, $p \in [0.3, 0.8]$, $k \in \{3, 4, 5\}$. *S6.* Complex Gaussian M with 3–5 blocks of size 2 and random block basis changes. *S7.* Random descriptors ($n_x \in [10, 30]$, $n_z \in [6, 20]$, $f_x = A + f_z g_z^{-1} g_x$) with five event classes. The relocation uses $\beta = \beta_2 = 1$ and the consistency pair (2, 0.5). *P1–P4.* Direct-path solutions (full AC equilibrium, central-difference Jacobians); P3 uses the common realization of Lemma 1. Seeds and rules are fixed in the preregistration file.

APPENDIX D

NONLINEAR SIMULATION DEFINITIONS

The TDS integrates the full DAE (BDF, relative tolerance 10^{-7} , maximum step 0.05 s), with Newton network solutions at every step. A 0.5-s disturbance is followed by 30 s of free response. $D(t)$ is the larger of the transverse state and voltage deviations, normalized by 10^{-3} and 10^{-2} .

- *Recovers:* D decays below 10% of its early peak with a negative envelope rate.
- *Fails:* D exceeds twice the early peak with positive growth over the last 10 s, or the relative rotor angle exceeds π .
- *Outside model scope:* a guard trips. The guards are bus voltage in $[0.8, 1.2]$; GFL terminal voltage in $[0.9, 1.1]$; GFL current or apparent power ≤ 1.2 ; reactive power ≤ 1.0 ; PLL or speed frequency in $[-3, 1.8]$ Hz; COI frequency within ± 0.5 Hz; field voltage within the documented exciter limits; and stabilizer output ≤ 0.1 (per unit on the device base).

Thresholds r_S come from continuation on a frozen amplitude grid followed by bisection. The minimum of the scope-censored thresholds defines ρ_{scope} .

REPRODUCIBILITY

The final scientific evidence state is frozen at commit b9f274e2. The post-freeze evidence addenda cited here are f775db89 (boundary anatomy), 7a772808/f2946257 (Monte Carlo preregistration and amendment), cfe9fe4b (Monte Carlo results), 5d0b1986/1f025cc3 (uncertainty-envelope preregistration and results), and 0a4e18c2, 7773ea8c (ANDES and pandapower reproductions); the ParaEMT EMT record is archived in the repository. Code, frozen configurations, and the source data of every figure are available from the author and in the project repository.

REFERENCES

- [1] N. Hatziaargyriou, J. Milanović, C. Rahmann, V. Ajarapu, C. Cañizares, I. Erlich, D. Hill, I. Hiskens, I. Kamwa, B. Pal, P. Pourbeik, J. Sanchez-Gasca, A. Stanković, T. Van Cutsem, V. Vittal, and C. Vournas, “Definition and classification of power system stability – revisited & extended,” *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3271–3281, 2021.
- [2] F. Milano, F. Dörfler, G. Hug, D. J. Hill, and G. Verbič, “Foundations and challenges of low-inertia systems,” in *Power Systems Computation Conference (PSCC)*, Dublin, Ireland, 2018.
- [3] U. Markovic, O. Stanojev, P. Aristidou, E. Vrettos, D. Callaway, and G. Hug, “Understanding small-signal stability of low-inertia systems,” *IEEE Transactions on Power Systems*, vol. 36, no. 5, pp. 3997–4017, 2021.
- [4] J. Sun, “Impedance-based stability criterion for grid-connected inverters,” *IEEE Transactions on Power Electronics*, vol. 26, no. 11, pp. 3075–3078, 2011.
- [5] L. Harnefors, M. Bongiorno, and S. Lundberg, “Input-admittance calculation and shaping for controlled voltage-source converters,” *IEEE Transactions on Industrial Electronics*, vol. 54, no. 6, pp. 3323–3334, 2007.
- [6] X. Wang and F. Blaabjerg, “Harmonic stability in power electronic-based power systems: Concept, modeling, and analysis,” *IEEE Transactions on Smart Grid*, vol. 10, no. 3, pp. 2858–2870, 2019.
- [7] H. Xin, W. Dong, X. Yuan, D. Gan, K. Wang, and H. Xie, “Generalized short circuit ratio for multi power electronic based devices infeed to power systems,” *Proceedings of the CSEE*, vol. 36, no. 22, pp. 6013–6027, 2016.

- [8] North American Electric Reliability Corporation, "Integrating inverter-based resources into low short circuit strength systems," NERC, Reliability Guideline, Dec. 2017.
- [9] A. G. J. MacFarlane, "Return-difference and return-ratio matrices and their use in analysis and design of multivariable feedback control systems," *Proceedings of the Institution of Electrical Engineers*, vol. 117, no. 10, pp. 2037–2049, 1970.
- [10] A. G. J. MacFarlane and I. Postlethwaite, "The generalized Nyquist stability criterion and multivariable root loci," *International Journal of Control*, vol. 25, no. 1, pp. 81–127, 1977.
- [11] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge University Press, 2012.
- [12] G.-C. Rota, "On the foundations of combinatorial theory I. Theory of Möbius functions," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, vol. 2, no. 4, pp. 340–368, 1964.
- [13] R. P. Stanley, *Combinatorics and Commutative Algebra*, 2nd ed. Boston: Birkhäuser, 1996.
- [14] D. Bienstock and A. Verma, "The $N - k$ problem in power grids: New models, formulations, and numerical experiments," *SIAM Journal on Optimization*, vol. 20, no. 5, pp. 2352–2380, 2010.
- [15] L. Farina and S. Rinaldi, *Positive Linear Systems: Theory and Applications*. New York: Wiley, 2000.
- [16] M. Magdon-Ismail, "NP-hardness and inapproximability of sparse PCA," *Information Processing Letters*, vol. 126, pp. 35–38, 2017.
- [17] P. W. Sauer and M. A. Pai, *Power System Dynamics and Stability*. Upper Saddle River, NJ: Prentice Hall, 1998.
- [18] R. M. Karp, "Reducibility among combinatorial problems," in *Complexity of Computer Computations*, R. E. Miller and J. W. Thatcher, Eds. New York: Plenum, 1972, pp. 85–103.
- [19] S. Poljak and J. Rohn, "Checking robust nonsingularity is NP-hard," *Mathematics of Control, Signals, and Systems*, vol. 6, no. 1, pp. 1–9, 1993.
- [20] A. Nemirovskii, "Several NP-hard problems arising in robust stability analysis," *Mathematics of Control, Signals, and Systems*, vol. 6, no. 2, pp. 99–105, 1993.
- [21] V. Katewa and F. Pasqualetti, "On the real stability radius of sparse systems," *Automatica*, vol. 113, p. 108685, 2020.
- [22] A. Brauer, "Limits for the characteristic roots of a matrix. IV: Applications to stochastic matrices," *Duke Mathematical Journal*, vol. 19, no. 1, pp. 75–91, 1952.
- [23] T. Athay, R. Podmore, and S. Virmani, "A practical method for the direct analysis of transient stability," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-98, no. 2, pp. 573–584, 1979.
- [24] H. Cui, F. Li, and K. Tomsovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [25] ANDES Development Team, "ANDES IEEE 39-bus test case," https://docs.andes.app/en/v1.9.3/getting_started/testcases/index.html, version 1.9.3 test-case index; dynamic case introduced in v1.8.7; accessed Aug. 28, 2026.
- [26] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed. New York: Springer, 2004.
- [27] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "pandapower—an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.
- [28] M. Xiong, B. Wang, D. Vaidhyanathan, J. Maack, M. Reynolds, A. Hoke, K. Sun, and J. Tan, "ParaEMT: An open source, parallelizable, and HPC-compatible EMT simulator for large-scale IBR-rich power grids," *IEEE Transactions on Power Delivery*, vol. 39, no. 2, pp. 911–921, 2024.
- [29] C. Cañizares *et al.*, "Benchmark systems for small-signal stability analysis and control," IEEE Power and Energy Society, Technical Report PES-TR18, 2015.
- [30] I. J. Pérez-Arriaga, G. C. Verghese, and F. C. Schweppe, "Selective modal analysis with applications to electric power systems, part I: Heuristic introduction," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-101, no. 9, pp. 3117–3125, 1982.
- [31] J. J. Sanchez-Gasca, V. Vittal, M. J. Gibbard, A. R. Messina, D. J. Vowles, S. Liu, and U. D. Annakkage, "Inclusion of higher order terms for small-signal (modal) analysis: Committee report – task force on assessing the need to include higher order terms for small-signal (modal) analysis," *IEEE Transactions on Power Systems*, vol. 20, no. 4, pp. 1886–1904, 2005.